

Results in Elementary Group Theory

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In this document we give essential results from elementary group theory. This text is not intended as a sole reference, but rather as a compact reference to someone with a background in abstract mathematics wishing to review the material.

Last Updated: October 24, 2017.

Hence, this text will not dwell on examples or definitions, but will rather focus on rigorous proofs for finite groups.

Part 1. Abstract Groups

1. Groups and Subgroups

Throughout this document, unless otherwise specified, we shall write G to denote a non-empty set and $\odot$ denotes an operation on G :

$$\odot : G \times G \longrightarrow G.$$

DEFINITION 1. A pair $(G, \odot)$ is called a group if

- (1) G is associative under $\odot$, i.e. if $a \odot (b \odot c) = (a \odot b) \odot c$ for all $a, b, c \in G$
- (2) There exists an element $e \in G$ such that $g \odot e = e \odot g = g$ for all $g \in G$. This is called the identity element of the group.
- (3) For each $g \in G$ there exists $g^{-1} \in G$ such that

$$g \odot g^{-1} = g^{-1} \odot g = e.$$

We will often omit the notation ' $\odot$ ' and simply write gh to mean $g \odot h$ for $g, h \in G$. Note that by our definition of $\odot$ it follows that $gh \in G$ whenever $g, h \in G$. Moreover, we shall write G to mean a group $(G, \odot)$ to simplify notation.

EXAMPLE 1. The set of real numbers $\mathbb{R}$ is a group when endowed with the usual notion of addition as the group operation. In fact, if $\mathcal{R}$ is a ring then the pair $(\mathcal{R}, +)$ is a group.

EXAMPLE 2. The set $\mathbb{C}^\times := \{z \in \mathbb{C} : z \neq 0\}$ is a group under multiplication.

EXAMPLE 3. Let $p \in \mathbb{N}$ be a prime; the set $(\mathbb{Z}/p\mathbb{Z})^\times$ is a group under multiplication.

We would first like to establish some consequences of the group axioms. First, this identity element is unique. Indeed, assume that e' satisfies condition (2) of our group definition. Then,

$$e = ef = fe = f.$$

In a similar vein, the inverse to any element is unique. To see that this is true fix $g \in G$ and assume $gh = hg = e$. Then,

$$g^{-1} = g^{-1}e = g^{-1}(gh) = (g^{-1}h)h = eh = h.$$

A group G is called **Abelian** if it is commutative, i.e. $gh = hg$ for all $g, h \in G$. We shall say that G has finite order if $\#G < \infty$.

DEFINITION 2. If G is a group, a set $H \subseteq G$ is called a subgroup of G whenever

- (1) $e \in H$.

- (2) Given $g, h \in H$ we have $gh \in H$.
 (3) For each $g \in H$ the inverse $g^{-1} \in H$.

From these axioms it is easy to verify that H is a group in its own right. The sets $\{e\}$ and G are called the trivial subgroups of G ; indeed they are always subgroups. We shall write $H < G$ as shorthand to say that H is a subgroup of G . A subgroup H is called **cyclic** provided there exists an element $g \in H$ such that

$$\langle g \rangle := \{g^m : m \in \mathbb{Z}\} = H.$$

In this case, we say that H is generated by g and that g is a generator of H . For any element $g \in H$ we then define the *order of h* , denoted $\text{ord}(g)$, to be the minimal positive integer k such that $g^k = e$. If no such element exists, we put $\text{ord}(g) = \infty$.

PROPOSITION 1.1. *Let G be a group and $g \in G$. Then $\text{ord}(h) = \#\langle g \rangle$.*

Proof. Suppose that $\text{ord}(h) = \infty$, then for any positive integers n and m we have $g^n \neq g^m$. Otherwise, assume without loss of generality that $m < n$ and note that g^m has inverse g^{-m} it follows that $g^{n-m} = e$ which contradicts $\text{ord}(g) = \infty$. Thus, the set $\langle g \rangle$ contains infinitely many distinct elements which gives $\#\langle g \rangle = \infty$.

Suppose now that $\text{ord}(g) = k < \infty$. Then, clearly,

$$\langle g \rangle = \{e, g, g^2, \dots, g^{\text{ord}(g)-1}\}$$

where all these elements must be distinct by the minimality of $\text{ord}(g)$. This concludes the proof. QED.

LEMMA 1.2. *Let G be a group and $\{H_\alpha\}_{\alpha \in I}$ an indexed family of subgroups of G .¹ Then $\bigcap_{\alpha \in I} H_\alpha$ is a subgroup of G .*

Proof. Since $e \in H_\alpha$ for all indices $\alpha \in I$ we have $e \in \bigcap_{\alpha \in I} H_\alpha$. Similarly, for each $g \in \bigcap_{\alpha \in I} H_\alpha$ we have $g \in H_\alpha$ for all α whence $g^{-1} \in H_\alpha$ for each $\alpha \in I$ which gives us $g^{-1} \in \bigcap_{\alpha \in I} H_\alpha$. A verbatim argument shows that $\bigcap_{\alpha \in I} H_\alpha$ is closed under the group operation of G . QED.

DEFINITION 3. Denote by G a group and let $\{g_\alpha\}_{\alpha \in I}$ a family of elements in G . The minimal subgroup generated by this family, which we denote $\langle \{g_\alpha\}_{\alpha \in I} \rangle$, is the group

$$\bigcap_{G > H \supseteq \{g_\alpha\}_{\alpha \in I}} H_\alpha.$$

¹Each H_α is a subgroup of G . Note that I need not be countable here!

2. Lagrange's Theorem for Groups

Lagrange's theorem is a simple, yet powerful result in the theory of finite groups. Indeed, it is a fundamental tool in the classification of low order groups. It has numerous applications in linear algebra, number theory and cryptography, as well. The theorem asserts the following:

THEOREM 2.1 (Lagrange's Theorem). *Let G be a finite group. If H is a subgroup of G , then $\#H$ divides $\#G$.*

2.1. Cosets. To prove Theorem 2.1 we require some preliminaries, namely the notion of a left coset in G . Let us fix a group G and let H be a subgroup of G . A left coset of H in G is a set of the form

$$gH := \{gh : h \in H\}$$

where $g \in G$ is fixed. Clearly, $gH \subseteq G$ by closure under the group operation. What makes this notion powerful is the following lemma.

LEMMA 2.2. *Define an equivalence relation $\sim$ on the group G by saying that $g \sim k$ if $g \in kH$. This is an equivalence relation such that $[g] = gH$.*

Proof. We first verify that this is indeed an equivalence relation on G . Since H is subgroup it contains the identity element which gives $g \in gH$, i.e. $g \sim g$. If $g \sim k$ we have $g = kh$ for some $h \in H$. This implies that $gh^{-1} = k$ whence $k \in gH$ since $h^{-1} \in H$. That is, we have $g \sim k$ if and only if $k \sim g$. Finally, suppose that $g \sim k$ and $k \sim m$. This is equivalent to saying that there are $h_{1,2} \in H$ such that

$$g = kh_1 \quad \text{and} \quad k = mh_2.$$

Putting this together gives $g = kh_1 = (mh_2)h_1 = m(h_2h_1) = mh'$, thereby yielding $g \in mH$ and $g \sim m$.

It remains only to show that $[g] = gH$ where $[g]$ is the equivalence class of g :

$$[g] = \{k \in G : g \sim k\}.$$

Let $k \in [g]$ be given, then $g \sim k$ which is equivalent to $k \sim g$. Thus, it follows that $k \in gH$ implying that $[g] \subseteq gH$. Conversely, let $k \in gH$. Then $k = gh$ for some $h \in H$. This is precisely the statement that $k \sim g$ which occurs if and only if $g \sim k$. Thus, $[g] = gH$ as was asserted. QED.

The above was tedious, but necessary.

PROPOSITION 2.3. *Let G be a group and H a subgroup of G . Then G is the disjoint union of equivalence classes, i.e.*

$$G = \bigsqcup_{g \in G} [g].$$

Proof. We first show $G = \bigcup_{g \in G} [g]$. Clearly, since $[g] \subseteq G$ it follows that $G \supseteq \bigcup_{g \in G} [g]$. To see the reverse inclusion, fix $g \in G$ and note that $g \in [g]$ since $g \sim g$ by our previous lemma. To complete the proof we need only show that given $g, h \in G$ we have either

$$[g] \cap [h] = \emptyset \quad \text{or} \quad [g] = [h].$$

Assume that $[g] \cap [h] \neq \emptyset$ and select ℓ from their intersection. Then $g \sim \ell$ and $k \sim \ell$. Given any $k \in G$ it follows that $k \in [h]$ since $k \sim g \sim \ell \sim k$. This implies that $[g] \subseteq [h]$. By symmetry, we also have $[h] \subseteq [g]$. QED.

We are now prepared to prove Lagrange's theorem, but shall first introduce some notation. Let G be a finite group and H a subgroup of G . We shall write $[G : H]$ to denote the number of disjoint equivalence classes in G (under the same equivalence relation as earlier).

Proof of Theorem 2.1. Using the proposition above it follows that we may write $G = \bigsqcup_{j=1}^n g_j H$, for some family of representatives² $\{g_j\}_{j \leq n}$ in G . We claim that $\#g_i H = \#g_j H$ for all suitable indices $i \neq j$.

Let $x, y \in G$. It suffices to find a bijection $f : xH \rightarrow yH$. We may assume without loss of generality that $x \neq y$. We do this by defining

$$f(g) := yx^{-1}g.$$

This is well defined since $g = xh$ whence $f(g) = yx^{-1}(xh) = yh$, which lives in yH . We first claim that this is injective. Assume that $f(a) = f(b)$ for $a, b \in xH$. This is to say that

$$yx^{-1}a = yx^{-1}b.$$

Since $yx^{-1} \in G$ it is invertible from which we obtain $a = b$. To see that this map is surjective, we pick $yH \ni b = yh$ for some h . We then take $a = xh$.

Returning now to our main argument, we have for any $g \in G$

$$\begin{aligned} \#G &= \# \left\{ \bigsqcup_{j=1}^n g_j H \right\} = \sum_{j=1}^n \# \{g_j H\} = \sum_{j=1}^n \# \{gH\} \\ &= n \# \{gH\}. \end{aligned}$$

This concludes the proof. QED.

COROLLARY 2.4. *Let G be a finite group and fix $g \in G$. Then $\text{ord}(g)$ divides $\#G$.*

²By a representative we mean that $g_j \in [g]$.

Proof. By our earlier remarks we have that $\#\langle g \rangle = \text{ord}(g)$; by Theorem 2.1 it is therefore sufficient to prove that $\langle g \rangle$ is a subgroup of G . Clearly, $e \in \langle g \rangle$ and by definition we have that $\langle g \rangle$ is closed under the group operation. If $h \in \langle g \rangle$ then $h = g^n$ for a minimal $n \in \mathbb{N}_0$ with $n < \text{ord}(g)$. It is now easy to check that

$$\langle g \rangle \ni g^{\text{ord}(g)-n} = g^{-1}.$$

QED.

COROLLARY 2.5. *Let G be a group with $\#G = p$ for some prime p . Then G is cyclic.*

Proof. Pick $g \in G$ with $g \neq e$. By the previous corollary we have $\#\langle g \rangle = \text{ord}(g) \mid \#G = p$. Since $g \neq e$ we have $\text{ord}(g) > 1$. Using that p is prime it follows that $p = \text{ord}(g)$ whence we obtain $\langle g \rangle = G$. QED.

3. Cyclic Groups

Given the final results from the previous section it makes sense to turn towards the study of cyclic groups for a little while. Cyclic groups are a beautiful topic with a great number of applications, especially in cryptography. Henceforth, for the remainder of this section, G will be a non-trivial cyclic group of order $n \in \mathbb{N}$.

LEMMA 3.1. *Let g be a generator of G and $1 \leq a \leq n$. Then*

$$\text{ord}(g^a) = \frac{n}{\gcd(a, n)}. \quad (3.1)$$

Proof. We first note that $g^t = e$ implies that $n \mid t$. Indeed, if this were not so then we would have the representation

$$t = s_1 n + s_2, \quad 0 < s_2 < n$$

whence we would obtain $g^t = g_1^s n g^{s_2} = g^{s_2} \neq e$ since $\text{ord}(g) = n$. Let $r \in \mathbb{N}$ be such that $n \mid ar$, it is clear that $(g^a)^r = e$. Moreover, $\text{ord}(g^a)$ must be the minimal such r . Otherwise, there exists an integer $r < \text{ord}(g^a)$ such that $n \mid ar$. However, we would have $(g^a)^r = e$ which would then contradict the definition of the order.

Therefore, we seek the minimal such r . Now, if $n \mid ar$, we could write

$$\left(\frac{n}{\gcd(a, n)} \right) \cdot \gcd(a, n) \mid ar,$$

where $\left(\frac{n}{\gcd(a, n)} \right)$ is co-prime to a . Thus, $\left(\frac{n}{\gcd(a, n)} \right) \mid r$. In particular, it divides the minimal such r . Thence, (3.1) holds true. QED.

COROLLARY 3.2. *Let G be a finite group of order n and g a generator of G , i.e. $\langle g \rangle = G$. Then $\langle g^a \rangle = G$ if and only if $\gcd(a, n) = 1$.*

Proof. $g \in G$ generates the group G if and only if $\text{ord}(g) = \#G$.

QED.

REMARK 1. This relates to the Euler totient function ϕ , which appears often in number theory and cryptography. Indeed, ϕ is a function defined by

$$\phi : \mathbb{N} \longrightarrow \mathbb{N}, \quad n \mapsto \#\{1 \leq k \leq n : \gcd(k, n) = 1\}.$$

Thus, a cyclic group G of order $n < \infty$ has $\phi(n)$ generators.

We have the following remarkable result for these cyclic groups of finite order.

THEOREM 3.3. *Let G be a finite cyclic group of order n . For each positive integer d dividing n the group G has a unique subgroup of order d . Furthermore, this subgroup is cyclic.*

Proof. This problem becomes quite a lot easier if we can show all subgroups of G are cyclic. Fix a subgroup H of G and assume without loss of generality that $H \neq \{0\}$ and $G \neq H$. Since $\langle g \rangle = G$ there exists a minimal $a \in \mathbb{N}$ such that $g^a \in H$. We claim this element generates this subgroup H . By closure under the group operation, $\langle g^a \rangle \subseteq H$. For the reverse inclusion, let $g^r \in H$. Of course, if $g^r = e$ we are done and we may thus assume that $r \geq 1$. By the division algorithm we may write

$$r = ak + m \quad \text{for } 0 \leq m < a.$$

By closure of H one has $g^{r-ak} = g^m \in H$. By minimality of a , we must have that $m = 0$. Thus, $g^r = g^{ak}$ which implies that $H = \langle g^a \rangle$.

Let d be as given, we shall now prove the existence of such a subgroup. By the previous lemma, for $1 \leq a \leq n$ there holds

$$\text{ord}(g^a) = \frac{n}{\gcd(a, n)}.$$

Since $d \mid n$ there exists $t \in \mathbb{N}$ such that $n = td$. Put $a = t$ whence

$$\text{ord}(g^t) = \frac{td}{\gcd(t, td)} = d.$$

It only remains to prove uniqueness. Consider the set

$$K := \{g \in G : g^d = e\}.$$

Clearly, K is a subgroup of G . Now let $H < G$ be order d . Then $H = \langle g^a \rangle$ where g is a generator of G and $\text{ord}(g^a) = d$. Hence, $H \subseteq K$. Conversely, let $x \in K$. This is

equivalent to writing

$$\begin{aligned}
& x = g^k \text{ and } g^{dk} = e \quad \text{with } 0 \leq k < n \\
& \iff x = g^k \text{ and } dk = nt \quad \text{with } 0 \leq k < n, t \in \mathbb{N}_0 \\
& \iff x = g^k \text{ and } k = \frac{nt}{d} \quad \text{with } 0 \leq k < n, t \in \mathbb{N}_0 \\
& \iff x = g^k \text{ and } k = \frac{nt}{d} \quad \text{with } 0 \leq k < n, 0 \leq t < d
\end{aligned}$$

whence $\#K \leq d$. Thus, $K = H$.

QED.

3.1. Applications to Finite Field Theory. Recall the definition of the Euler totient function ϕ over $\mathbb{N}$.

LEMMA 3.4. *Let $n \in \mathbb{N}$, then*

$$n = \sum_{d|n} \phi(d). \quad (3.2)$$

Proof. Let G be a cyclic group of order n .³ Then, counting the elements of each order and using that fact that the order of an element divides the order of the group:

$$n = |G| = \sum_{d|n} \#\{g \in G : \text{ord}(g) = d\}.$$

Let $d \mid n$, by our previous theorem there exists a unique $H < G$ of order d , generated by each element of order d . By an earlier remark we know that there are exactly $\phi(d)$ elements in G of this order. Hence,

$$n = \sum_{d|n} \#\{g \in G : \text{ord}(g) = d\} = \sum_{d|n} \phi(d).$$

QED.

LEMMA 3.5. *Let G be a group of order n such that for each $d \mid n$ there exists at-most one subgroup of order d . Then G is cyclic.*

Proof. We proceed in two steps.

Step 1. If there exists an element $g \in G$ of order d there are exactly $\phi(d)$ elements of order d in G .

³Such a group always exists. Indeed, let $n \in \mathbb{N}$ and observe that $(\mathbb{Z}/n\mathbb{Z}, +)$ is a group of order n .

Proof of Step 1. The set $\langle g \rangle$ is by assumption the only subgroup of order d in G , and is a cyclic group in its own right. Moreover, by this uniqueness, any element of order d in G must live in $\langle g \rangle$. Since this is a cyclic group, it follows that it has exactly $\phi(d)$ generators.

Step 2. G is cyclic.

Proof of Step 2. Clearly,

$$n = \sum_{d|n} \#\{g \in G : \text{ord}(g) = d\} = \sum_{d|n} \varepsilon_d \phi(d)$$

where $\varepsilon_d = 0$ if no element of order d exists in G and 1 otherwise. Now, since $n = \sum_{d|n} \phi(d)$ by our previous lemma we conclude that $\varepsilon_n = 1$. Hence, G is cyclic.

QED.

We apply it to give the following stupendous result for finite fields. This result is a generalization of the primitive root theorem, which states that for any prime p the group $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic.

COROLLARY 3.6. *Let $\mathbb{F}$ be a finite field. The group of units $\mathbb{F}^\times$ is cyclic.*

Proof. Let $n := \#\mathbb{F}$ and assume $d \mid n - 1$. Let H be a subgroup of order d ; to apply the previous lemma we must show that H is unique. Indeed, for each $x \in H$ we would have $x^d = 1$ whence the elements of H belong to the set

$$\{x \in \mathbb{F} : x^d - 1 = 0\}.$$

We know that this set has at most d elements, and so it follows that H is unique.

QED.

4. Normal and Quotient Groups

In this section we let G denote an arbitrary group, unless stated otherwise.

DEFINITION 4. Let H be a subgroup of G . We say that H is **normal** provided $gH = Hg$ for all $g \in G$. In this case we write $H \triangleleft G$.

We seek to give the following characterization of such subgroups:

PROPOSITION 4.1. *Let G be a group and N a subgroup of G . The following are equivalent:*

- (1) $N \triangleleft G$.
- (2) $gN \subseteq Ng$ for all $g \in G$.
- (3) $gng^{-1} \in N$ for all $n \in N$ and $g \in G$.

Proof. (1 $\implies$ 2). Assume that $N \triangleleft G$, then $gN = Ng$ for all $g \in G$ which gives (2).

(2 $\implies$ 3). Assume that $gN \subseteq Ng$. Let $g \in G$ and $n \in N$ so that $gn \in gN$. In particular, $gn = n'g$ for some $n' \in N$. Hence, $gng^{-1} = n' \in N$.

(3 $\implies$ 1). We note that our hypothesis implies that $gN \subseteq Ng$ for all $g \in G$. Hence, to establish (1) it suffices to prove that $Ng \subseteq gN$. Let $x = ng \in Ng$ be given. Then write

$$x = ng = gg^{-1}ng = gg^{-1}ng = gn' \in gN.$$

QED.

We are now capable of defining **quotient groups**. Let G be a group and $N \triangleleft G$. We denote by G/N the collection of sets gN with $g \in G$. Of course, we must now give this set a group structure. Any two elements $x, y \in G/N$ take the form

$$x = aN, \quad y = bN$$

with $a, b \in G$. We then define $\odot : x \mapsto (ab)N$. We claim that this is well defined, i.e. if $aN = a'N$ and $bN = b'N$ then $(ab)N = (a'b')N$. To see that this is so we argue as follows. First observe that $aN = a'N$ implies that $a' = an_1$ and, similarly, $b' = bn_2$ where $n_{1,2} \in N$. Thus,

$$(a'b')N = an_1bn_2N = abb^{-1}n_1bn_2N = abn'_1n_2N = ab\tilde{n}N = abN$$

since $kN = N$ whenever $k \in N$. In case this is not clear, we give a proof. Since N is closed under the group operation it is obvious that $kN \subseteq N$. Given $n \in N$ note that $n = kk^{-1}n = k(k^{-1}n) \in kN$.

It is now easy for the reader to check that the identity of G/N is eN and that the inverse of aN is $a^{-1}N$.

LEMMA 4.2. *Let B and N be subgroups of G with $N \triangleleft G$.*

- (1) $B \cap N \triangleleft B$.
- (2) BN is a subgroup of G . Moreover, $BN = NB$.
- (3) If $B \triangleleft G$ then $BN \triangleleft G$ and $B \cap N \triangleleft G$.
- (4) If B and N are finite

$$\#BN = \frac{\#B \cdot \#N}{\#(B \cap N)}.$$

Proof. We first prove (1). In light of the previous proposition, it suffices to show that $bnb^{-1} \in B \cap N$ for all $b \in B$ and $n \in B \cap N$. First note that $B \cap N$ is a subgroup of G and therefore of B . Now

$$bnb^{-1} \in N$$

since $N \triangleleft G$. To see that $bnb^{-1} \in B$ as well, we need only note that $n \in B \cap N$ implies $n \in B$. Thus, the result follows by the subgroup property of B .

We now prove (2). Let $x = bn \in BN$ and $y = b'n' \in BN$. Write

$$xy = bnb'n' = bb'([b']^{-1}nb')n' = bb'\tilde{n}n' \in BN.$$

Clearly, the identity $e = ee \in BN$ and the inverse of bn is precisely $n^{-1}b^{-1} \in BN$. Since N is a normal subgroup:

$$BN = \bigcup_{b \in B} bN = \bigcup_{b \in B} Nb = NB.$$

Let us now prove (3). We begin by showing that $BN \triangleleft G$. That is, we show that for $k \in BN$ one has $gkg^{-1} \in BN$ for all $g \in G$. Certainly, we write since $N \triangleleft G$

$$gkg^{-1} = gbn g^{-1} = gb g^{-1} (ng^{-1}) = gb g^{-1} \tilde{n} = \tilde{b}\tilde{n} \in BN.$$

As for $B \cap N \triangleleft G$, we show that $gkg^{-1} \in B \cap N$ for all $g \in G$ and $k \in B \cap N$. Indeed,

$$gkg^{-1} \in B$$

and

$$gkg^{-1} \in N$$

since $B, N \triangleleft G$. We now prove (4). We consider a surjective map:

$$f : B \times N \longrightarrow BN, \quad (b, n) \mapsto bn.$$

This is clearly well defined and surjective. Thus, it suffices to show that for each $x \in BN$ the fibre $f^{-1}(x)$ is a set of cardinality $\#(B \cap N)$. Indeed, fix $x = bn \in BN$ and consider the set

$$\mathcal{X} := \{(by, y^{-1}n) : y \in B \cap N\}$$

which consists of $\#(B \cap N)$ elements. If $y = (by, y^{-1}n) \in \mathcal{X}$ then $f(y) = bn = x$ so that $\mathcal{X} \subseteq f^{-1}(x)$. To see the reverse inclusion we fix $z = (b_1, n_1) \in f^{-1}(x)$ so that $f(z) = b_1 n_1 = bn$. Now, define

$$y = b^{-1}b_1 = nn_1^{-1} \in B \cap N.$$

Then, $z = (b_1, n_1) = (by, y^{-1}n) \in \mathcal{X}$. This concludes the proof. QED.

5. Homomorphisms of Groups

In this section we study maps between groups that preserve the structure of each group. More precisely, if G and H are groups, a mapping

$$f : G \longrightarrow H$$

is called a **homomorphism** if $f(ab) = f(a)f(b)$ for all $a, b \in G$. In particular, $f(e) = e$ since $f(e) = f(e \cdot e) = f(e)f(e)$ so that $f(e) = e$. It also follows that $f(a^{-1}) = f(a)^{-1}$. To see that this is so write

$$f(a)f(a^{-1}) = f(a \cdot a^{-1}) = f(e) = e.$$

A homomorphism of groups is called an **isomorphism** if it is a bijection. Given a homomorphism of groups f , we define the **kernel** of this homomorphism by

$$\text{Ker } f := \{g \in G : f(g) = e\}.$$

PROPOSITION 5.1. *Let $f : G \longrightarrow H$ be a homomorphism of groups. Then $\text{Ker } f \triangleleft G$ and f is injective if and only if $\text{Ker } f = \{e\}$. Moreover, for $h \in \text{Im } f$, the fibre $f^{-1}(h)$ is a coset of the kernel.*

Proof. We first verify that $\text{Ker } f$ is a subgroup of G . By our earlier observations it is clear that $e \in \text{Ker } f$. If $a, b \in \text{Ker } f$ we then have $f(ab) = f(a)f(b) = ee = e$ whence $ab \in \text{Ker } f$. Moreover, $f(a^{-1}) = f(a)^{-1} = e^{-1} = e$. Thus, $\text{Ker } f < G$.

We now show that $\text{Ker } f \triangleleft G$. It suffices to show that $gng^{-1} \in \text{Ker } f$ for all $n \in \text{Ker } f$ and $g \in G$. Certainly, this follows immediately from the following

$$f(gng^{-1}) = f(g)f(n)f(g^{-1}) = f(g)f(g^{-1}) = f(gg^{-1}) = f(e) = e.$$

Hence, $\text{Ker } f \triangleleft G$. Assume that f is injective; since $f(e) = e$ it follows that $\text{Ker } f = \{e\}$. Conversely, assume that the kernel of f is trivial. If $f(a) = f(b)$ then $e = f(a)f(b)^{-1} = f(ab^{-1})$ which implies that $ab^{-1} \in \text{Ker } f$. In particular, $ab^{-1} = e$ which gives $a = b$.

Assume that $h \in \text{Im } f$, i.e. there exists $g \in G$ such that $f(g) = h$. We claim that the fibre $f^{-1}(h)$ is the coset $g\text{Ker } f$. Assume that $x \in g\text{Ker } f$ so that $x = gn$ with $n \in \text{Ker } f$, then $f(x) = f(gn) = f(g)f(n) = f(g) = h$. That is, $g\text{Ker } f \subseteq f^{-1}(h)$. To see the reverse inclusion, take $y \in f^{-1}(h)$ and observe that $f(y) = f(g)$. It follows that $f(yg^{-1}) = e$ whence $yg^{-1} \in \text{Ker } f$ so that $y \in g\text{Ker } f$. The proof is now complete. QED.

LEMMA 5.2. *Let G be a group and $N \triangleleft G$. The canonical map*

$$\pi_N : G \longrightarrow G/N, \quad g \mapsto gN,$$

is a surjective homomorphism of groups. Moreover, $\text{Ker } \pi_N = N$.

Proof. We first verify that π_N is a homomorphism. Clearly, given any two $a, b \in G$,

$$\pi_N(ab) = (ab)N = \pi_N(a)\pi_N(b).$$

To see that π_N is a surjection, recall that an element of G/N is a coset of the form gN for some $g \in G$. Then $\pi_N(g) = gN$ as required. Let $n \in N$, we claim that $nN = N$. Since N is a subgroup of G , it is immediate that $nN \subseteq N$. For the reverse

inclusion fix $k \in N$ and write $k = n(n^{-1}k) \in nN$. This implies that $N \subseteq \text{Ker } \pi_N$. Conversely let $x \in \text{Ker } \pi_N$. Then it follows that $\pi_N(y) = eN = N$. We are done if we can show that $y \in N$. We proceed by contradiction, suppose that $y \notin N$. Then

$$\pi_N(y) = yN \neq N$$

which follows from the fact that $y \in yN$ (since $e \in N$).

QED.

5.1. Homomorphisms acting on subgroups. We again assume that G and H are groups.

PROPOSITION 5.3. *Let A be a subgroup of G and $f : G \longrightarrow H$ a homomorphism of groups.*

- (1) $f(A)$ is a subgroup of H .
- (2) If B is a subgroup of H the pre-image $f^{-1}(B)$ is a subgroup of G .
- (3) If $B \triangleleft H$ one has $f^{-1}(B) \triangleleft G$.
- (4) If f is surjective and $A \triangleleft G$ it follows that $f(A) \triangleleft H$.

Proof. Since $e \in A$ we obtain $f(e) = e \in f(A)$. Let $x, y \in f(A)$ and take $a, b \in A$ such that $f(a) = x$ and $f(b) = y$. Then, $xy = f(a)f(b) = f(ab)$. Since $ab \in A$ it follows that $xy \in f(A)$. Finally, if $x = f(a)$ we have that $x^{-1} = f(a)^{-1} = f(a^{-1}) \in f(A)$ since $a^{-1} \in A$. This verifies (1).

Assume that $B < H$. Since $e \in B$ $f^{-1}(B)$ is non-empty and contains the identity. Let $a, b \in f^{-1}(B)$ be given. $f(ab) = f(a)f(b)$ where $f(a), f(b) \in B$. Thus, $f(ab) \in B$ whence $ab \in f^{-1}(B)$. Finally, take $a \in f^{-1}(B)$ and observe that $f(a^{-1}) = f(a)^{-1} \in B$ by assumption. We then conclude that $f^{-1}(B)$ is a subgroup of G .

Let $B \triangleleft H$, we must show that $gng^{-1} \in f^{-1}(B)$ for all $g \in G$ and $n \in f^{-1}(B)$. Certainly, we note that

$$f(gng^{-1}) = f(g)f(n)f(g^{-1}) \in B$$

since $B \triangleleft H$ and $f(n) \in B$. This verifies (3). Now, to show (4) we assume that f is also a surjection. We shall show that $hkh^{-1} \in f(A)$ whenever $h \in H$ and $k \in f(A)$. First let us take $g \in G$ and $n \in A$ such that $f(g) = h$ and $f(n) = k$. We note that $gng^{-1} \in A$ since $A \triangleleft G$. Now,

$$f(gng^{-1}) = f(g)f(n)f(g^{-1}) = hkh^{-1} \in f(A).$$

QED.

Now, we present a very useful lemma.

LEMMA 5.4. *Let G be a group H a subgroup of G . Then $gH = H$ if and only if $g \in H$.*

Proof. Let $g \in H$. Then by closure under the group operation we find $gH \subseteq H$. To see the reverse inclusion fix $h \in H$. Then $h = gg^{-1}h = g\tilde{h} \in gH$. Conversely, assume that $gH = H$. Since $e \in H$ it follows that $g \in gH = H$. QED.

5.2. Classification of Prime Groups. Let p be a prime and G be a finite group of order p . We know there exists a group of order p (consider $\mathbb{Z}/p\mathbb{Z}$ under addition) and that this group is also cyclic (1 is a generator). Here we shall show that all groups of order p are cyclic and isomorphic.

THEOREM 5.5. *Let p be a prime and G a finite group of order p . Then G is cyclic and is unique up to isomorphism.*

Proof. It is clear that $G \supsetneq \{e\}$ and so there exists an element $x \neq e$ in G . This will generate the group. Indeed, $\langle x \rangle$ is a subgroup of G with $\#\langle x \rangle > 1$ and, by Lagrange's theorem, we know $\#\langle x \rangle \mid \#G = p$. Since p is prime it follows that $\text{ord}(x) = p$.

We now prove that G is unique up to isomorphism. Suppose H is any other group of order p ; we know from the above that H is cyclic. Let x be a generator of G and y a generator of H . Define a mapping

$$\psi : G \longrightarrow H, \quad x^a \mapsto y^a.$$

This is a well defined map. Indeed, $x^a = x^b$ if and only if $b = pk + a$ for some $k \in \mathbb{Z}$ so that

$$\psi(x^b) = y^b = y^{a+pk} = y^a \cdot e = \psi(x^a)$$

since $\text{ord}(y) = p$. To see that ψ is a homomorphism of groups we fix $\alpha, \beta \in G$ and write

$$\alpha = x^a, \quad \beta = x^b$$

for positive integers a, b . Now,

$$\psi(\alpha\beta) = \psi(x^{a+b}) = y^{a+b} = y^a y^b = \psi(\alpha)\psi(\beta).$$

Since G and H are finite and of the same cardinality, it suffices to show that ψ is injective. Suppose $\alpha \in G$ satisfies $\psi(\alpha) = e$. Write $\alpha = x^a$ for some $a \in \mathbb{N}$ and note that $y^a = e$. Since y has order p this implies that $p \mid a$. This implies that $x^a = e$ so that $\text{Ker } \psi = \{e\}$. QED.

6. The First Isomorphism Theorem

In this section we present the following fundametal result. Yet again, we let G and H be groups. Two groups G_1 and G_2 are said to be **isomorphic**, written $G_1 \cong G_2$, if there exists a bijective homomorphism $\varphi : G_1 \longrightarrow G_2$.

THEOREM 6.1. *Let $f : G \rightarrow H$ be a homomorphism of groups. There exists an injective homomorphism $f' : G/\text{Ker } f \rightarrow H$ such that the following diagram commutes⁴:*

$$\begin{array}{ccc} G & \xrightarrow{f} & H \\ & \searrow \pi_{\text{Ker } f} & \nearrow f' \\ & G/\text{Ker } f & \end{array} \quad (\text{FI})$$

Epecially, $G/\text{Ker } f \cong f(G)$.

Proof. We define $f' : G/\text{Ker } f \rightarrow H$ by taking $g\text{Ker } f \mapsto f(g) \in H$. We first show that this is well defined, i.e. that if $a\text{Ker } f = b\text{Ker } f$ then $f(a) = f(b)$. Certainly, if $a\text{Ker } f = b\text{Ker } f$ then $b^{-1}a \in \text{Ker } f$ so that

$$f(a) = f(bb^{-1}a) = f(b)f(e) = f(b)$$

as required. Now, we need to prove that f' is a homomorphism. This is easy:

$$f'([g_1\text{Ker } f][g_2\text{Ker } f]) = f(g_1g_2) = f(g_1)f(g_2).$$

Now, to show that f' is injective we argue as follows. Assume

$$f'(a\text{Ker } f) = f(a) = f'(b\text{Ker } f)f(b).$$

Then $f(ab^{-1}) = e$ so that $ab^{-1} \in \text{Ker } f$, or, rather $a \in b\text{Ker } f$. Thus, if $x \in a\text{Ker } f$ we have $x = ah$ where $h \in \text{Ker } f$. But then

$$x = ah = bh' \in b\text{Ker } f, \quad h' \in \text{Ker } f.$$

Therefore, $a\text{Ker } f \subseteq b\text{Ker } f$. By symmetry we have equality. QED.

REMARK 2. In (FI) we can write $f = f' \circ \pi_{\text{Ker } f}$ which (by surjectivity of $\pi_{\text{Ker } f}$) shows that $f'(G/\text{Ker } f) = f(G)$. Since f' is an injective homomorphism, it follows that $G/\text{Ker } f \cong f(G)$.

7. The Second Isomorphism Theorem

THEOREM 7.1. *Let G be a group and fix two subgroups B, N of G . If $N \triangleleft G$ there holds $B/(B \cap N) \cong BN/N$.*

Proof. We must first check that all is well in the claim. By a previous result we know that $B \cap N \triangleleft B$ so that the quotient group $B/(B \cap N)$ makes sense. We need only check that $N \triangleleft BN$. Indeed, let $b \in BN$ and $n \in N$, one has

$$bnb^{-1} \in N$$

⁴In this case, by commutativity, we mean that $f = f' \circ \pi_{\text{Ker } f}$.

since $b \in G$ as well. Now we define

$$f : B/(B \cap N) \longrightarrow BN/N, \quad b(B \cap N) \mapsto bN = beN.$$

We must check that this is well defined. Assume $a(B \cap N) = b(B \cap N)$. Then $an_1 = bn_2$ where $n_{1,2} \in B \cap N$. That is, we may write $b = an$ for some $n \in B \cap N$. Then,

$$bN = anN = aN.$$

Clearly, f is a homomorphism of groups and is surjective; given an element $bnN = bN \in BN/N$ take $x = b(B \cap N) \in B/(B \cap N)$ as a point mapping to bnN under the action of f . Finally, we must show that f is injective. Suppose that

$$f(a(B \cap N)) = aN = bN = f(b(B \cap N)).$$

Since $aN = bN$ it follows that we may write $b^{-1}a = n$, for some $n \in N$. Note that since $a, b \in B$ we also have $n \in N$. Now,

$$a(B \cap N) = bb^{-1}a(B \cap N) = bn(B \cap N) = b(B \cap N).$$

Thus, f is an isomorphism and the proof is complete. QED.

8. The Third Isomorphism Theorem

We dedicate this section to the proof of the third isomorphism theorem for groups. Mostly, this isomorphism theorem is an application of the first isomorphism theorem.

THEOREM 8.1. *Let G be a group and assume $H, K \triangleleft G$ with $H < K$. Then*

$$(G/K) \cong (G/H)/(K/H).$$

Proof. We check that the statement makes sense. Certainly, since $H, K \triangleleft G$ it is clear that the quotient groups G/K and G/H are well defined. Now, if $h \in H$ and $k \in K$ we have

$$khk^{-1} \in H$$

since $H \triangleleft G$ and $k \in G$. Therefore $H \triangleleft K$ and K/H is well defined. We now define a map

$$f : G/H \longrightarrow G/K, \quad gH \mapsto gK.$$

We must, as always, check that this function is well defined. Assume that $aH = bH$ with $a, b \in G$; we claim that $aK = bK$. Indeed, this first equality tells us that $b = ah$ for some $h \in H$. Hence,

$$bK = ahK = aK$$

since $h \in H < K$. Now, it is obvious from the definition that f is a homomorphism of groups. It is surjective because given $gK \in G/K$ we have

$$gH \xrightarrow{f} gK.$$

Now, we wish to determine the kernel of f . Observe that

$$\text{Ker } f = \{gH \in G/H : g \in G, gK = K\} = \{gH \in G/H : g \in G, g \in K\} = K/H.$$

Therefore, by the first isomorphism theorem,

$$f(G/H) = G/K \cong (G/H)/\text{Ker } f = (G/H)/(K/H).$$

This completes the proof. QED.

The third isomorphism theorem has a famous ‘companion’, which is often referred to as the correspondence theorem.

THEOREM 8.2 (Correspondence Theorem). *Let $f : G \longrightarrow H$ be a surjective homomorphism of groups. There exists a bijection*

$$F : \{\text{subgps. of } G \text{ containing } \text{Ker } f\} \longrightarrow \{\text{subgps. of } H\}.$$

Proof. We introduce notation. For the remainder of the proof we denote by G_1 an arbitrary subgroup of G containing the normal subgroup $\text{Ker } f$ of G ; H_1 will denote an arbitrary subgroup of H . We then define

$$F(G_1) := f(G_1) < H.$$

By earlier results, it is indeed true that $f(G_1)$ is a subgroup of H . We also construct an auxiliary function:

$$F'(H_1) := f^{-1}(H_1) < G.$$

It suffices to show that $F \circ F' = \mathbf{1}$ and $F' \circ F = \mathbf{1}$. First, fix $H_1 < H$. Let $y \in H_1$. Then $y = f(x)$ for some $x \in f^{-1}(H_1)$. It follows that $y = f(x) \in f(f^{-1}(H_1))$, i.e. $H_1 \subseteq f(f^{-1}(H_1))$. If $y \in f(f^{-1}(H_1))$ then there exists $x \in f^{-1}(H_1)$ so that $y = f(x)$ whence $f(x) \in H_1$. This shows that $H_1 = f(f^{-1}(H_1))$. Hence,

$$F \circ F' = \mathbf{1}.$$

Let $G_1 < G$ be as above. Let $x \in G_1$ so that $f(x) \in f(G_1)$. It follows that $x \in f^{-1}(f(G_1))$ whence $G_1 \subseteq f^{-1}(f(G_1))$. For the reverse (and final) inclusion we fix $x \in f^{-1}(f(G_1))$ so that $f(x) \in f(G_1)$. Take $g \in G_1$ so that $f(x) = f(g)$. Since f is a homomorphism of groups it follows that $xg^{-1} \in \text{Ker } f$. Hence, $x \in [\text{Ker } f]g = g\text{Ker } f \in G_1$ where this last inclusion follows from the fact that $g \in G_1$ and $\text{Ker } f < G_1$. This shows that $F' \circ F = \mathbf{1}$ which completes the proof of the correspondence theorem. QED.

9. Group Actions

Throughout this section we shall fix a group G and a non-empty set S . We say that G acts upon S if we are given a function

$$\star : G \times S \longrightarrow S, \quad (g, s) \mapsto g \star s$$

that satisfies:

- (1) $e \star s = s$ for all $s \in S$,
- (2) $(g_1 g_2) \star s = (g_1) \star (g_2 \star s)$ for all $s \in S$ and $g_1, g_2 \in G$.

Given an action on (G, S) we can defined two new sets. For each $s \in S$ we define the **orbit** and **stabilizer** of s , respectively, by

$$\text{Orb}(s) := \{g \star s : g \in G\} \subseteq S$$

and

$$\text{Stab}(s) := \{g \in G : g \star s = s\} \subseteq G.$$

LEMMA 9.1.

- (1) We shall say that $s_1 \sim s_2$ if $g \star s_1 = s_2$ for some $g \in G$. This is an equivalence relation on S and $[s_1] = \text{Orb}(s_1)$.
- (2) $\text{Orb}(s_1)$ is a subgroup of G .
- (3) If both S and G are finite in cardinality,

$$|\text{Orb}(s)| = \frac{|G|}{|\text{Stab}(s)|}.$$

Proof. We begin by verifying that ' $\sim$ ' is indeed an equivalence relation on S . Clearly, we can write $s_1 = e \star s$ for all $s \in S$ which implies that $s \sim s$. Assume $s_1 \sim s_2$, then $g \star s_1 = s_2$ for some $g \in G$. We then have

$$g^{-1} \star (g \star s_1) = (g^{-1} g) \star s_1 = s_1 = g^{-1} \star s_2$$

whence $s_2 \sim s_1$. Finally, let $s_1 \sim s_2$ and $s_2 \sim s_3$. We have for suitable $g_1, g_2 \in G$:

$$g_1 \star s_1 = s_2, \quad g_2 \star s_2 = s_3.$$

In particular we have $s_2 = g_2^{-1} \star s_3$ so that

$$g_1 \star s_1 = s_2 = g_2^{-1} \star s_3$$

which implies that $s_1 \sim s_3$. We shall now show that $\text{Stab}(s)$ is a subgroup of G . Clearly, $e \in \text{Stab}(s)$. If $g \in \text{Stab}(s)$ then

$$g \star s = s$$

so that $g^{-1} \star (g \star s) = e \star s = s = g^{-1} \star s$. This implies that $g^{-1} \in \text{Stab}(s)$. Finally, if $g_1, g_2 \in \text{Stab}(s)$ then $g_1 g_2 \in \text{Stab}(s)$ since

$$(g_1 g_2) \star s = g_1 \star (g_2 \star s) = g_1 \star s = s.$$

Now, since $\text{Stab}(s)$ is a subgroup of G it suffices (by Lagrange's theorem) to show that there exists a bijection

$$\text{Orb}(s) \leftrightarrow \{\text{left cosets of } \text{Stab}(s) \text{ in } G\}.$$

To ease notation, we define $H := \text{Stab}(s)$ and define

$$\varphi(gH) := g * s.$$

We claim first that this is well defined. Assume $g_1H = g_2H$ for $(g_1, g_2) \in G \times G$. It follows that $g_2^{-1}g_1 = h \in H$. Thence,

$$\varphi(g_1H) = g_1 * s = (eg_1) * s = (g_2g_2^{-1}g_1) * s = g_2 * (h * s) = g_2 * s.$$

Clearly, φ is surjective since any element $g * s$ in the orbit of s can be mapped to by gH . Finally, to establish that φ is an injection, assume

$$\varphi(g_1H) = g_1 * s = \varphi(g_2H) = g_2 * s.$$

Since $g_1 * s = g_2 * s$ it follows that $g_2^{-1}g_1 * s = s$ whence $g_2^{-1}g_1 \in H = \text{Stab}(s)$. Thus,

$$g_1H = g_2g_2^{-1}g_1H = g_2H$$

as was required. QED.

REMARK 3. Suppose that we are given a group G and a set S such that G acts upon S . By the lemma above, it follows that S is the disjoint union of orbits in S . Indeed, this follows from the argument used in the proof of Lagrange's theorem which shows that equivalence classes (i.e. orbits) are either disjoint or equal.

9.1. Cayley's Isomorphism Theorem. In this section we realize finite groups as subgroups of permutation groups. More precisely, it will be shown that a group G of order $n \in \mathbb{N}$ is isomorphic to a subgroup of S_n , the permutation group on n -letters.

Before we can prove this, we must introduce some notation. If A is a finite set of size n we shall denote by Σ_A the set of all bijections $f : A \rightarrow A$. It is not difficult to show that Σ_A is then a group under composition. It is then clear that we may associate Σ_A to S_n , i.e.

$$\Sigma_A \cong S_n.$$

LEMMA 9.2. *Let G be a group and S a non-empty set. A group action of G upon S induces a homomorphism of groups*

$$\Psi : G \rightarrow \Sigma_S.$$

Conversely, any homomorphism of this form will generate a group action on S .

Proof. Assume we are given an action of G upon S . We define a mapping

$$\phi_g : S \longrightarrow S, \quad \phi_g(s) = g * s$$

and subsequently define $\Psi(g) := \phi_g \in \Sigma_S$. We show that this notion is well defined, i.e. that ϕ_g is indeed a bijection. If $g * s = g * \tilde{s}$ for $s, \tilde{s} \in S$ then the invertibility of g implies that $s = \tilde{s}$. This establishes injectivity. To see that ϕ_g is surjective, we fix $s \in S$. Clearly $e * s = s$ so that

$$s = (gg^{-1}) * s = g * (g^{-1} * s) = g * \tilde{s}.$$

Hence, Ψ is well defined. It remains to prove that Ψ is a homomorphism of groups, in its own right. Clearly, $\Psi(e)$ is the identity map, as required. Now, if $g, h \in G$ we find that for each $s \in S$

$$\Psi(gh) = \phi_{gh}(s) = (gh) * s = g * (h * s) = g * (\phi_h(s)) = (\phi_g \circ \phi_h)(s).$$

Conversely, assume we are given a map Ψ as above. We define an action of G on S by taking

$$G \times S \ni (g, s) \mapsto \Psi(g)(s).$$

QED.

THEOREM 9.3 (Cayley). *Let G be a group of order $n \in \mathbb{N}$. Then G is isomorphic to a subgroup of S_n .*

Proof. We consider a particular action of G upon itself. Let $S = G$ and define

$$\star : G \times G \longrightarrow G, \quad (g_1, g_2) \mapsto g_1 g_2.$$

Clearly, this is a well defined action. By the previous lemma, this action induces a homomorphism

$$\Psi : G \longrightarrow \Sigma_n, \quad \Psi(g) = \phi_g.$$

We claim that Ψ is injective. Indeed, assume that $\Psi(g) = \Psi(h)$ for $g, h \in G$. Then

$$\phi_g(s) = g * s = h * s = \phi_h(s)$$

for all $s \in S = G$. In particular, $ge = he$ so that $g = h$. This concludes the proof.

QED.

9.2. The Coset Representations. Let G be a group and suppose $H < G$ is a subgroup of finite index, say, $[G : H] = n$. We consider the action of G upon the left-cosets of H in G :

$$\star : G \times G/H, \quad (x, gH) \mapsto xgH$$

where we fix a collection of representatives for G/H . By virtue of the previous discussions, this is equivalent to giving a homomorphism:

$$\Psi : G \longrightarrow \Sigma_{G/H}, \quad g \mapsto \phi_g$$

where $\phi_g(xH) = gxH$. We now consider $\text{Ker } \Psi$. This subgroup of G is precisely the collection:

$$\begin{aligned} \{g \in G : \phi_g(xH) = xH, \forall xH \in G/H\} &= \bigcap_{xH \in G/H} \{g \in G : \phi_g(xH) = xH\} \\ &= \bigcap_{x \in G} \{g \in G : gxH = xH\}. \end{aligned}$$

Now, $gxH = xH$ if and only if $x^{-1}gxH = H$ which is equivalent to saying that $x^{-1}gx \in H$. Therefore,

$$\text{Ker } \Psi = \bigcap_{x \in G} \{g \in G : gxH = xH\} = \bigcap_{x \in G} xHx^{-1}.$$

PROPOSITION 9.4. *Let G be a finite group and H a subgroup of index p in G where p is the least prime dividing $\#G$. Then $H \triangleleft G$.*

Proof. We define an action of G upon the left cosets of H in the natural way. Certainly, define

$$\star : G \times G/H, \quad (x, gH) \mapsto xgH$$

where we fix representatives for the cosets of H in G . This then allows us to define a homomorphism

$$\psi : G \longrightarrow S_p$$

since $[G : H] = p$. Let $K := \text{Ker } \psi$ and observe that K is normal in G . Thus, by virtue of the first isomorphism theorem, G/K is isomorphic to a subgroup of S_p . Now, $[G : K]$ must also divide G which implies that $[G : K] = p$. Finally,

$$p = [G : H] = [G : K] \cdot [H : K] = [H : K] \cdot p$$

which shows that $[H : K] = 1$ whence $H = K$.

QED.

10. Burnside's Lemma and Cauchy's Theorem

Let G be a group of finite order acting upon a non-empty set S . Given an element $g \in G$ we define

$$I(g) := \#\{s \in S : g * s = s\}.$$

That is, $I(g)$ is the number of elements in S fixed by the action of G .

THEOREM 10.1 (Burnside). *Let N denote the number of orbits of G in S . There holds*

$$N = \frac{1}{|G|} \sum_{g \in G} I(g). \quad (10.1)$$

Proof. We first define a function

$$T : G \times S \longrightarrow \{0, 1\}, \quad (g, s) \mapsto \begin{cases} 1 & \text{if } g * s = s, \\ 0 & \text{else.} \end{cases}$$

Since we may write S as the disjoint union of orbits, we may choose representatives $s_1, \dots, s_N$ for these equivalence classes whence

$$\begin{aligned} \sum_{g \in G} I(g) &= \sum_{g \in G} \sum_{s \in S} T(g, s) = \sum_{s \in S} \sum_{g \in G} T(g, s) = \sum_{s \in S} |\text{Stab}(s)| \\ &= \sum_{s \in S} \frac{|G|}{|\text{Orb}(s)|} \\ &= \sum_{j=1}^N \sum_{s \in \text{Orb}(s_j)} \frac{|G|}{|\text{Orb}(s_j)|} \\ &= \sum_{j=1}^N |G| = N |G|. \end{aligned}$$

This concludes the proof.

QED.

We shall now conclude this part by proving Cauchy's Theorem for groups. The proof is somewhat messy and there are a lot of “moving parts” to keep track of. The advantage, however, is that the proof is elementary and uses few complicated or deep results.

THEOREM 10.2 (Cauchy). *Let G be a group of order n and let p be a prime dividing n . There exists an element of order p in G .*

Proof. We shall assume that $n > 2$, since the case $n = 2$ is obvious from the fact that only e has order 1. Now, let T be the set of all p -tuples in G , i.e. let

$$T := \prod_{j=1}^p G.$$

We then denote by S the set of all elements in T upto a cyclic shift by an element of $\mathbb{Z}/p\mathbb{Z}$. Note that in fact S is a set of equivalence classes and not a set of p -tuples. The first step is to count the elements of S .

Step 1. We define now an action of $\mathbb{Z}/p\mathbb{Z}$ on T by taking a number x modulo p and a p -tuple t which we map to a right cyclic shift of t by p . Then S is also the

set of Orbits in T under this action so that, by Burnside's theorem,

$$\#S = \frac{n^p - n}{p} + n.$$

Step 2. We now prove the existence of an element with order p . Observe that $p \mid n$ implies that $p \nmid \#S$ by the expression found in Step 1. Now, define an action

$$\star : G \times S \longrightarrow S, \quad g(g_1, g_2, \dots, g_p) := (gg_1, gg_2, \dots, gg_p)$$

where $(g_1, g_2, \dots, g_p)$ is some representative of a class in G . This is well defined since the image of any two representatives will differ by a cyclic permutation at most. Since S is the disjoint union of orbits, it follows that at-least one orbit has size less than n . That is,

$$\frac{\#G}{\#\text{Stab}(s)} < n$$

for some $s \in S$. Hence, the stabilizer of this s is non-trivial and we can extract $e \neq g \in \text{Stab}(s)$. Then, for some associated p -tuple:

$$(g_1, g_2, \dots, g_p) = (gg_1, gg_2, \dots, gg_p) = (g_{i+1}, g_{i+2}, \dots, g_p, g_1, \dots, g_i).$$

Therefore $gg_1 = g_{i+1}$, $g^2g_1 = gg_{i+1} = g_{2i+1}$ and so forth. Therefore,

$$g^p g_1 = g_{pi+1}$$

whence $g^p = e$. Since p is prime and $g \neq e$ it follows that $\text{ord}(g) = p$. QED.

Part 2. Important Classes of Groups

Fix $n \in \mathbb{N}$ and let S_n denote the group of permutations on n -letters. It is not too difficult to prove that any permutation $\sigma \in S_n$ can be written as the product of disjoint cycles:

$$\sigma = \sigma_1 \sigma_2 \cdots \sigma_r.$$

It is obvious that disjoint cycles commute. Given a cycle $(t_1, \dots, t_k)$ we write $|(t_1, \dots, t_k)| = k$ to denote the length of the cycle. We may assume that

$$|\sigma_1| \leq |\sigma_2| \leq \cdots \leq |\sigma_r|.$$

We will also allow the cycles of length 1. It then follows that

$$n = |\sigma_1| + \cdots + |\sigma_r|.$$

11. Conjugation Classes and The Class Equation

In this section we fix a finite group G and let H be a subgroup of G . We shall denote by $[G : H]$ the index of H in G , i.e. $\#G/\#H$. There is a natural action of G upon itself given by

$$\star : G \times G \longrightarrow G, \quad (g, x) \mapsto g \star x = gxg^{-1}.$$

It is left as a simple exercise to verify that this is indeed a group action upon G . Now, we define for $x \in G$:

$$\text{Conj}(x) := \text{Orb}(x) = \{g \star x : g \in G\} = \{gxg^{-1} : g \in G\}$$

and

$$C(x) := \text{Stab}(x) = \{g \in G : g \star x = x\} = \{g \in G : gx = xg\}.$$

By earlier results we know that $C(x)$ is a subgroup of G and that for all $x \in G$ there exists a bijection

$$G/C(x) \longrightarrow \text{Conj}(x).^5$$

Furthermore, G is the disjoint union of orbits which implies that $G = \bigsqcup_{x \text{ repr.}} \text{Orb}(x)$ whence we obtain

$$\#G = \sum_{x \text{ repr.}} \#\text{Conj}(x) = \sum_{x \text{ repr.}} \frac{\#G}{\#C(x)}.$$

Let $Z(G) := \{x \in G : gx = xg, \forall g \in G\}$ and observe that if $x \in Z(G)$ there holds trivially

$$\text{Conj}(x) = \{x\}.$$

Hence, it follows that

$$\#G = \#Z(G) + \sum_{G \setminus Z(G) \ni x \text{ repr.}} \frac{\#G}{\#C(x)}.$$

This is known as the class equation.

11.1. Counting Finite Groups. For the remainder of this section we study groups of finite order. Given $n \in \mathbb{N}$ we shall show that, up to isomorphism, there are only finitely many groups of order n .

PROPOSITION 11.1. *Fix $n \in \mathbb{N}$. There are only finitely many groups of order n up to isomorphism.*

Proof. If G is a set consisting of n elements there can only be finitely many functions $G \times G \longrightarrow G$. QED.

⁵In fact, the proof for this fact did not require G to be a finite group!

LEMMA 11.2. Let $q \in \mathbb{Q}$ with $q > 0$. Given $t \in \mathbb{N}$ there exist at most finitely many n_j , $1 \leq j \leq t$, such that

$$q = \frac{1}{n_1} + \frac{1}{n_2} + \cdots + \frac{1}{n_t}.$$

Proof. We argue by induction on t . If we take $t = 1$ then the claim is trivial. Assume the statement holds true for $t - 1$. Suppose that one can write

$$q = \sum_{j=1}^t \frac{1}{n_j}$$

for $n_t \in \mathbb{N}$. We may assume that $n_1 \geq n_2 \geq \cdots \geq n_t$ so that

$$q \leq \sum_{j=1}^t \frac{1}{n_t} = \frac{t}{n_t}$$

whence we find that $n_t \leq t/q$. This implies that there are only finitely many possible choices of n_t . Now, if we apply the induction hypothesis to $q - \frac{1}{n_t}$ the claim follows. QED.

This may be applied to give an easy proof of the following theorem.

THEOREM 11.3. Let $d \in \mathbb{N}$, there exists at most finitely many groups (of finite order) with d conjugacy classes.

Proof. Assume there exists a group G with d conjugacy classes. Recall the class equation:

$$\#G = \#Z(G) + \sum_{G \setminus Z(G) \ni x \text{ repr.}} \frac{\#G}{\#C(x)}$$

which implies that

$$1 = \frac{1}{\#G} + \cdots + \frac{1}{\#G} + \frac{1}{C(x_1)} + \cdots + \frac{1}{C(x_d)}$$

where we have exactly d terms on the RHS. Since there are at most finitely many ways to write this, it follows that we can find at most finitely many such groups by virtue of the previous lemma and proposition. QED.

12. The Simplicity of A_n

Let $n \in \mathbb{N}$ and let S_n denote the group of permutations on n -letters. If G is a group we shall say that G is **simple** provided the only normal subgroups of G are $\{e\}$ and G itself. This section is devoted to showing that the alternating group A_n

on n -letters (the subgroup of S_4 consisting of even permutations) is simple for $n \geq 5$.⁶

LEMMA 12.1. *Let G be a group and H a subgroup of G . Then H is normal in G if and only if H is the disjoint union of conjugacy classes.*

Proof. We have the subsequent equivalency:

$$H \triangleleft G \iff ghg^{-1} \in H, \quad \forall g \in G, h \in H \iff \text{Conj}(h) \subseteq H, \quad \forall h \in H.$$

Now, assume $H \triangleleft G$. For all $h \in H$ we find that $\text{Conj}(h) \subseteq H$ so that

$$\bigsqcup_{h \text{ repr.}} \text{Conj}(h) = H$$

since $h \in \text{Conj}(h)$. Conversely, assume we have an index set Λ so that

$$H = \bigsqcup_{\lambda \in \Lambda} \text{Conj}(h_\lambda)$$

where $h_\lambda \in H$. Let now $h \in H$ be given and note that $h \in \text{Conj}(h_\lambda)$ for some $\lambda \in \Lambda$. Clearly, $\text{Conj}(h) = \text{Conj}(h_\lambda)$ so that $H \triangleleft G$ by the statement above. QED.

THEOREM 12.2. *The alternating group A_n is simple for $n \geq 5$.*

Proof. We omit this proof. QED.

13. p -Groups

In this section we consider finite groups of a particular type. Henceforth, we shall say a finite group G is a p -group if p is prime and there exists $n \in \mathbb{N}$ so that $\#G = p^n$.

LEMMA 13.1. *Let G be a p -group, the center of G is non-trivial.*

Proof. Recall that the center of G is $Z(G) := \{x \in G : gx = xg, \forall g \in G\}$. We may obviously assume without harm that $Z(G) \subsetneq G$. Then $Z(G)$ is a proper subgroup of G and there exists $x \in G$ with $x \notin Z(G)$. Since $x \notin Z(G)$, it follows that $C(x)$ is a proper subgroup G and therefore

$$p \mid \frac{\#G}{\#C(x)}.$$

⁶Recall that any permutation $\sigma \in S_n$ can be written as the product of transpositions. The sign of σ , written $\text{sgn}(\sigma)$, is the number of such transpositions. In fact, if $(a_1, \dots, a_\ell)$ is a cycle, we have $\text{sgn}((a_1, \dots, a_\ell)) = (-1)^{\ell-1}$.

This holds for all $x \notin Z(G)$ whence, by the class equation,

$$\#Z(G) = \#G - \sum_{\substack{x \notin Z(G) \\ x \text{ repr.}}} \frac{\#G}{\#C(x)}.$$

This implies that $p \mid \#Z(G)$. Especially, $\#Z(G) \geq p$.

QED.

THEOREM 13.2. *Let G be a p -group.*

- (1) *For each proper subgroup $H \triangleleft G$ there exists a normal subgroup K in G such that $H < K < G$ and $[K : H] = p$.*
- (2) *There exists a finite sequence of subgroups*

$$\{e\} =: H_0 < H_1 < \cdots < H_n = G$$

such that $\#H_j = p^j$ for each $j = 0, 1, \dots, n$.

Proof. We have two statements to prove.

- (1) Since H is a normal subgroup of G we know that the collection G/H is a quotient group. By Lagrange's theorem we know that $[G : H]$ is a power of p and therefore that G/H is a p -group in its own right. By the previous lemma, we know that $Z(G/H)$ is non-trivial. Choose $x \in Z(G/H)$ different from the identity. It then follows that $\text{ord}(x)$ is a power of p , say, a . Define then

$$y := x^{p^{a-1}}$$

so that $\text{ord}(y) = p$. Set $W := \langle y \rangle \triangleleft G/H$. Then W has order p . Recall that the canonical projection

$$\pi_H : G \longrightarrow G/H, \quad g \mapsto gH$$

is a surjective homomorphism of groups. Define $K := \pi_H^{-1}(W)$ and observe that $H = \text{Ker } \pi_H \subseteq K$. By the third isomorphism theorem,

$$K/H \cong W$$

which implies that $[K : H] = p$, as required.

- (2) We proceed by induction on j . Clearly, $H_0 < G$ is a proper normal subgroup. By the first part we can select a normal subgroup $H_1 \triangleleft G$ such that $[H_1 : H_0] = p$. Of course, this is equivalent (by Lagrange's theorem) to writing

$$\frac{\#H_1}{\#H_0} = \#H_1 = p.$$

Assume the case for j , if $j = n$ then we are done. We may then suppose that $j < n$. Then we have a subgroup $H_j \triangleleft G$. Repeating the first part, we

may select a normal subgroup $H_{j+1} \supseteq H_j$ such that $[H_{j+1} : H_j] = p$. The result follows by noting that

$$\frac{\#H_{j+1}}{\#H_j} = [H_{j+1} : H_j] = p.$$

QED.

LEMMA 13.3. *Let G be a group and suppose $H \triangleleft Z(G)$. If G/H is cyclic, G is Abelian.*

Proof. We remark that it suffices to assume that $H < Z(G)$. Now, G/H is cyclic and is therefore generated by some element xH where $x \in G$. Let $g_1, g_2 \in G$ be given and express them as

$$g_1 = x^i h_1, \quad g_2 = x^j h_2.$$

Indeed, this follows from the fact that G is the disjoint union of left cosets. Since the elements of H commute with those of G :

$$g_1 g_2 = (x^i h_1)(x^j h_2) = (x^j h_2)(x^i h_1) = g_1 g_2.$$

QED.

In the previous part of these notes we gave a rather imaginative proof of Cauchy's theorem. Below we give a simpler (and easier to understand) proof of Cauchy's theorem for Abelian groups. Indeed, it is less powerful but will be sufficient in many cases.

THEOREM 13.4 (Cauchy's Theorem—Take 2). *Let G be a finite Abelian group and p a prime dividing the order of G . Then there exists an element of order p in G .*

Proof. We proceed by induction on $\#G$. The case $\#G = 2$ is trivial since G is cyclic (and therefore has an element of order 2). Now assume that the claim holds for all groups of order strictly less than $\#G$. Let N be a maximal proper subgroup of G ; we shall distinguish two cases. If $p \mid \#N$ then we need only apply our induction hypothesis to conclude. Hence, we may assume that $p \nmid \#N$. Since N is a proper subgroup of G we may pick $x \in G \setminus N$ and define

$$B := \langle x \rangle \Delta G.$$

Indeed, B is normal in G since G is Abelian. Now consider the product group BN which is a subgroup of G strictly containing N . Thus, $BN = G$ which implies that

$$p \mid \#G = \#BN = \frac{\#B \cdot \#N}{\#(B \cap N)}.$$

Since $p \nmid \#N$ it follows that $p \mid \#B = \text{ord}(x)$. Write

$$\text{ord}(x) = pa$$

for some a . Then $\text{ord}(x^a) = p$ and we are done.

QED.

14. Sylow's Theorems

This section is devoted to the proofs of Sylow's theorems. We fix a group G of finite order and let $p \mid \#G$ be a prime with $\#G = p^r m$ for $r \in \mathbb{N}$ such that $\gcd(p^r, m) = 1$. We recall that given a subgroup H in G we define

$$N_G(H) := \{g \in G : gHg^{-1} = H\}.$$

It is easy to show that $N_G(H)$ is in fact a subgroup of G .

PROPOSITION 14.1. *There exists a subgroup of order p^r in G .*

Proof. We argue by induction on the order of G . If $\#G = 2$ the claim is clear as G is in fact cyclic. Assume now that the claim holds true up to, but not including, the order of G . Assume first that $p \mid \#Z(G)$. Then, since $Z(G)$ is in fact Abelian we may apply the previous lemma to extract an element $x \in Z(G)$ with $\text{ord}(x) = p$. Now consider the subgroup

$$N := \langle x \rangle \triangleleft Z(G).$$

We also note that $N \triangleleft G$ since elements of N commute with those in G . Consider now the canonical project map:

$$\pi_N : G \longrightarrow G/N, \quad g \mapsto gN.$$

Note that $[G : N] = p^{r-1}m$ and so, by our induction hypothesis, there exists a subgroup H' of order p^{r-1} in G/N . Now, by the correspondence theorem, we may select a subgroup

$$N = \text{Ker } \pi_N \triangleleft H < G$$

such that $\pi_N(H) = H'$. An application of the third isomorphism theorem then yields that

$$H/N \cong \pi_N(H)/\pi_N(N) = H'/\{e_{G/N}\} = H'.$$

Hence, $\#H = \#N \cdot \#H' = p^r$, as was required. Now we handle the case where $p \nmid \#Z(G)$. By the class equation we may write

$$\#G = \#Z(G) + \sum_{\substack{x \text{ repr.} \\ x \notin Z(G)}} \frac{\#G}{C(x)}$$

where $C(x) := \{g \in G : gx = xg\}$. Note that by assumption there exists $x \notin Z(G)$ such that $p \nmid \frac{\#G}{C(x)}$. This implies that $\#C(x) \geq p^r$. Moreover, since $x \notin Z(G)$ we conclude that $C(x)$ is a proper subgroup of G . Then, by our induction hypothesis we are done. QED.

LEMMA 14.2. *Let P be a maximal p -subgroup of G and let Q be any p -subgroup of G . Then*

$$P \cap Q = Q \cap N_G(P).$$

Proof. Note first that $P \subseteq N_G(P) = \{g \in G : gPg^{-1} = P\}$ whence $P \cap Q \subseteq Q \cap N_G(P)$. Define

$$A := Q \cap N_G(P)$$

and note that A is a p -subgroup of $N_G(P)$ (and therefore of G). By construction it is clear that $P \triangleleft N_G(P)$ so that, in fact, AP is a subgroup of $N_G(P)$. However,

$$\#(AP) = \frac{\#A \cdot \#P}{\#(A \cap P)}.$$

Then, it is clear that $p \mid \#AP$ which implies that AP is a p -group containing P . It follows that $AP = P$. In particular,

$$A \subseteq P$$

which implies that $A \subseteq P \cap Q$.

QED.

We now have the necessary tools to prove Sylow's theorem.

THEOREM 14.3 (Sylow). *Every maximal p -subgroup has order p^r and such a subgroup exists⁷. All these subgroups are conjugates and the number n_p of such subgroups satisfies*

- (i) $n_p \mid m$,
- (ii) $n_p \equiv 1 \pmod{p}$.

Proof. Fix a Sylow subgroup of G , say, P . We know that such a subgroup exists by virtue of the first proposition in this section. Define

$$S := \{P_1, P_2, \dots, P_a\}$$

where there P_j 's are conjugate to $P = P_1$, i.e.

$$S = \{gPg^{-1} : g \in G\}.$$

It will be proven in the exercises that a map $P \longrightarrow gPg^{-1}$ given by $p \mapsto gpg^{-1}$ is in fact an isomorphism of groups. This implies, in particular, that each P_j is a maximal p -subgroup.

Let now Q be any p -subgroup of G . We define an action of Q upon S by conjugation. That is, we construct a natural map

$$(g, P_j) \mapsto gP_jg^{-1}.$$

⁷Such a subgroup is called a Sylow p -subgroup

Then

$$\# \text{Orb}(P_j) = \frac{\#Q}{\# \text{Stab}(P_j)}$$

where $\text{Stab}(P_j) = \{q \in Q : qP_jq^{-1} = P_j\} = Q \cap N_G(P_j) = Q \cap P_j$. Therefore, if $P_j \subseteq Q$ we have that

$$\text{Orb}(P_j) = 1$$

and a proper power of p otherwise (since $Q \cap P_j$ is a subgroup of Q). We now consider the particular action of P_1 on S . By grace of these remarks, we have that

$$\# \text{Orb}(P_1) = 1$$

and that $\# \text{Orb}(P_j)$ is a power of p for all other j (since P_1 is not contained in any other p -group). Since S is the union of disjoint orbits, we have that

$$a = \#S = 1 + tp, \quad t \in \mathbb{N}_0.$$

This shows that $a \equiv 1 \pmod{p}$. Note that

$$\text{Orb}(P_1) = S$$

so that

$$a = \# \text{Orb}(P_1) = \frac{\#G}{\# \text{Stab}(P_1)} = \frac{\#G}{\#N_G(P_1)}.$$

In particular, $a \mid \#G$ which implies $a \mid m$. It now remains only to show that any maximal p -subgroup is in S . To this end, fix $Q < G$ a maximal p -subgroup. We shall show that it is conjugate to $P = P_1$. Otherwise, $Q \neq P_i$ for all $i = 1, \dots, a$ and so, by our earlier remarks, $p \mid a$: contradiction. QED.

Part 3. Exercises and Solutions

In this part we solve certain problems in group theory, mostly taken from assignments in Math 456 (Honours Algebra 3) at McGill University. We mostly stick to the problems that admit elegant solutions and omit those with tedious or technical answers.

EXERCISE 1. A group G is said to be a **boolean group** provided $g^2 = e$ for all $g \in G$. Show that a boolean group G is Abelian and that $\#G = 2^n$ for some $n \in \mathbb{N}$.

Solution. The first observation is that $g^{-1} = g$ for all $g \in G$. Indeed, this follows immediately from the fact that $g^2 = e$. In particular $(gh)^{-1} = gh$. Therefore

$$gh = (gh)^{-1} = h^{-1}g^{-1} = hg.$$

Now, assume p is a prime dividing $\#G$ (we are assuming $G \neq \{e\}$) and suppose that $p \geq 3$. By Cauchy's theorem for Abelian groups we could extract an element

$g \in G$ of order p . This is impossible since $g^2 = e$. Therefore $p \mid \#G$ implies $p = 2$ which concludes the proof. QED.

EXERCISE 2. Let $\mathbb{F}$ and $\mathbb{K}$ be finite fields and assume that $\mathbb{F} \subseteq \mathbb{K}$. Prove that $\#\mathbb{K} = q^n$ where $q := \#\mathbb{F}$.

Proof. The “trick” is to view $\mathbb{K}$ as a vector space over $\mathbb{F}$. Indeed, it is not difficult to see that $\mathbb{K}$ is a vector space over $\mathbb{F}$ when endowed with the natural notions of addition and scalar multiplication. Moreover, all of these will be well defined since $\mathbb{F} \subseteq \mathbb{K}$. Since $\mathbb{K}$ is a finite field, one must have $\dim(\mathbb{K}) = d < \infty$.

Let us now fix a basis $\{b_j\}_{j=1}^d$ and note that a vector $u \in \mathbb{K}$ is completely determined by its representation:

$$u = \alpha_1 b_1 + \cdots + \alpha_d b_d, \quad \alpha_j \in \mathbb{F}.$$

By counting the number of d -tuples $(\alpha_1, \alpha_2, \dots, \alpha_d)$ we find that $\#\mathbb{K} = q^d$ as was required. QED.

EXERCISE 3. In the context of the previous exercise, prove that if $a^q = a$ for $a \in \mathbb{K}$, then $a \in \mathbb{F}$.

Solution. Assume $a^q = a$. If $a = 0$ then the result is trivially true (recall that a field contains the additive identity). Hence, we need only consider the particular case $a^{q-1} = 1$ for $a \in \mathbb{F}^\times$.

Let now $x \in \mathbb{K}^\times$ be an element of order d where $d \mid \#\mathbb{F}^\times$ and $d \mid \#\mathbb{K}^\times$. We claim that $x \in \mathbb{F}^\times$. To see that this is so, observe that $\langle x \rangle$ is a subgroup of order d in $\mathbb{K}^\times$ and therefore it is unique. Since $\mathbb{F}^\times$ is also cyclic and $d \mid \#\mathbb{F}^\times$, there exists a unique subgroup $H < \mathbb{F}^\times$ of order d . Since $H < \mathbb{K}^\times$ it follows that $H = \langle x \rangle$ so that $x \in \mathbb{F}^\times$.

Finally note that $\text{ord}(a) \mid q - 1 \mid q^n - 1$. We now need only apply the first part to deduce that $a \in \mathbb{F}^\times$. QED.

EXERCISE 4. Let G and H be groups of finite order and let

$$a := \#G, \quad b := \#H.$$

If $\gcd(a, b) = 1$ then every homomorphism $f : G \longrightarrow H$ is trivial.

Solution. If $a = 1$ or $b = 1$ then the claim is trivially true since any such homomorphism takes e_G to e_H . We can therefore assume that $a, b \geq 2$. We proceed by contradiction; assume that f is non-trivial. Then, $\text{Ker } f \triangleleft G$ is a proper subgroup. This implies that

$$[G : \text{Ker } f] = \frac{\#G}{\#\text{Ker } f} > 1$$

is a divisor of $\#G$. Now, by the first isomorphism theorem,

$$G/\text{Ker } f \cong f(G)$$

where $f(G) < H$ has order $[G : \text{Ker } f]$. However this implies that

$$\gcd(a, b) \geq [G : \text{Ker } f] > 1$$

which is a contradiction.

QED.

14.1. Group Automorphisms. In the problems that follow we prove facts regarding automorphisms of groups, rather than make them a part of the notes. We used some of these results in the proof of Sylow's theorems.

We recall that, given a group G , a map $f : G \rightarrow G$ is called an automorphism provided it is an isomorphism. The set of all automorphisms is denoted $\text{Aut}(G)$.

EXERCISE 5. Let G be a group and P a subgroup of G . Prove that for each fixed $g \in G$ the map

$$\psi : P \rightarrow gPg^{-1}, \quad p \mapsto gpg^{-1}$$

is an isomorphism of groups.

Solution. We first verify that gPg^{-1} is indeed a group. Clearly, $e \in gPg^{-1}$ and for

$$x_1 = gp_1g^{-1}, \quad x_2 = gp_2g^{-1}$$

there holds $x_1x_2 = gp_1p_2g^{-1} \in gPg^{-1}$. Also,

$$(gp_1g^{-1})gp_1^{-1}g^{-1} = e.$$

Now, it is clear from the above that ψ is a homomorphism. Obviously, ψ is surjective. Assume now that

$$gp_1g^{-1} = gp_2g^{-1},$$

since g and g^{-1} are invertible we must have $p_1 = p_2$ which proves that ψ is indeed an isomorphism.

QED.

EXERCISE 6. In this problem we give $\text{Aut}(G)$ a group structure. Fix a group G and prove each of the following:

- (i) $\text{Aut}(G)$ is a group under composition.
- (ii) For a fixed $g \in G$ the mapping

$$\tau_g : G \rightarrow G, \quad x \mapsto gxg^{-1}$$

is an element of $\text{Aut}(G)$. Furthermore,

$$\tau : G \rightarrow \text{Aut}(G), \quad g \mapsto \tau_g$$

is a homomorphism of groups.

- (iii) $\text{Ker } \tau = Z(G)$, the center of G .

Solution. We first prove (i). Let f, g be homomorphisms from G into G , we claim that $f \circ g$ is again a homomorphism. Certainly, if $a, b \in G$ then

$$(f \circ g)(ab) = f(g(a)g(b)) = f(g(a))f(g(b)) = (f \circ g)(a) \cdot (f \circ g)(b).$$

Now, since the composition of bijections is a bijection it follows that $\text{Aut}(G)$ is closed under composition. The identity map is clearly $f = \mathbb{1}_G$ which takes an element $g \rightarrow g$. Obviously, $f \in \text{Aut}(G)$ has an inverse function f' that is again a bijection; we need only prove f' is a homomorphism $G \rightarrow G$.

Fix $a, b \in G$ and note that since f is a bijection

$$\begin{aligned} f^{-1}(ab) = f^{-1}(a)f^{-1}(b) &\iff f(f^{-1}(ab)) = f(f^{-1}(a)f^{-1}(b)) \\ &\iff f(f^{-1}(ab)) = f(f^{-1}(a))f(f^{-1}(b)) \\ &\iff ab = ab. \end{aligned}$$

Thus f' is a homomorphism of groups.

Now we establish (ii). Fix g and note that

$$\tau_g(a)\tau_g(b) = (gag^{-1})(gbg^{-1}) = gabg^{-1} = \tau_g(ab).$$

Clearly, if $\tau_g(a) = \tau_g(b)$ then $gag^{-1} = gbg^{-1}$ so that $a = b$. Thus τ_g is an injective homomorphism between two finite sets; it follows that τ_g is an isomorphism and thus lives in $\text{Aut}(G)$. Now to show that $g \mapsto \tau_g$ is a homomorphism of groups we fix $g, h \in G$ and let $a \in G$. There holds

$$\tau_{gh}(a) = (gh)a(gh)^{-1} = ghah^{-1}g^{-1} = (\tau_g \circ \tau_h)(a)$$

where $a \in G$ is arbitrary.

To establish (iii) we must show that $Z(G) = \text{Ker } \tau$. Clearly, if $Z(G) \ni g$ then for all $a \in G$

$$\tau_g(a) = gag^{-1} = agg^{-1} = a$$

which shows that $\tau_g = \mathbb{1}_G$. Hence, $Z(G) \subseteq \text{Ker } \tau$. For the reverse inclusion we find that if $g \in \text{Ker } \tau$ the map τ_g is given by $\tau_g(a) = a$ for all $a \in G$. Therefore,

$$a = gag^{-1} \iff ga = ag$$

for all $a \in G$.

QED.

14.2. Group Actions & Applications.

EXERCISE 7. Let G be a group acting on a non-empty set S and suppose $\sigma \in \text{Orb}(s)$ for $s \in S$. Show that $\text{Stab}(\sigma)$ is conjugate to $\text{Stab}(s)$, i.e.

$$\text{Stab}(\sigma) = g \text{Stab}(s) g^{-1}$$

for some $g \in G$.

Solution. Since $\sigma \in \text{Orb}(s)$ there exists $g \in G$ such that $\sigma = g * s$, where $*$ denotes the action G upon S . We claim that

$$\text{Stab}(\sigma) = g \text{Stab}(s) g^{-1}.$$

Assume $x \in \text{Stab}(\sigma)$, then $\sigma = x * \sigma$. Note that

$$(g^{-1} x g) * s = (g^{-1} x) * (g * s) = g^{-1} * (x * \sigma) = g^{-1} * \sigma = s.$$

This implies that $g^{-1} x g \in \text{Stab}(s)$ which yields $\text{Stab}(\sigma) \subseteq \text{Stab}(s)$. For the reverse inclusion we pick $x \in G$ such that $x * s = s$ and write

$$(g x g^{-1}) * \sigma = (g x) * (g^{-1} * \sigma) = g * (x * s) = g * s = \sigma.$$

Hence we have equality, i.e. $\text{Stab}(\sigma) = \text{Stab}(s)$.

QED.

EXERCISE 8. Let H and K be two subgroups of G and assume both H and K are of finite index.⁸ Prove that $H \cap K$ is a subgroup of finite index.

Solution. Define $S := (G/H) \times (G/K)$; we know by our assumption that S consists of finitely many elements. We shall now define an action of G upon S by letting

$$g * (g_1 H, g_2 K) \mapsto (g g_1 H, g g_2 K)$$

where we fix a set of representatives for G/H and G/K . This is clearly well defined. We now consider $\text{Stab}((H, K))$. This stabilizer is given by:

$$\{g \in G : gH = H, gK = K\} = H \cap K.$$

Now, the proof of Lemma 9.1 shows that there exists a bijection:

$$\{\text{left cosets of } H \cap K \text{ in } G\} \longleftrightarrow \text{Orb}((H, K)).$$

It is readily seen that

$$\text{Orb}((H, K)) = \{(gH, gK) : g \in G\}$$

whence we clearly have

$$\#\text{Orb}((H, K)) \leq \#(G/H \times G/K) < \infty.$$

Therefore, $[G : (H \cap K)]$ is also finite.

QED.

⁸We mean that both $[G : H]$ and $[G : K]$ are finite.

EXERCISE 9. We say a group G is **simple** if the only normal subgroups in G are $\{e\}$ and G . Let G be a simple group of order n . Show that if a proper subgroup H has index $k \neq 1$ then $k! \geq n$.

Proof. We define an action of G upon the left cosets of H in G , which we denote S .⁹ We set

$$g * (g_1 H) := (gg_1)H;$$

this is clearly well defined. Write

$$S := \{g_1 H, g_2 H, \dots, g_k H\}$$

and let g act on the whole set S by taking S to

$$\{gg_1 H, gg_2 H, \dots, gg_k H\}.$$

Now,

$$gg_1 H = gg_2 H \iff g_1 H = g_2 H$$

which shows that this procedure returns a permutation of the original set S . In this way we can associate a homomorphism of groups $\psi : G \longrightarrow S_k$. By previous results, we know that $\text{Ker } \psi \triangleleft G$ which shows that $\text{Ker } \psi = \{e\}$ or $\text{Ker } \psi = G$. In the first case, ψ is injective which implies that

$$\#G \leq \#S_k = k!.$$

If $\text{Ker } \psi = G$ then $\psi(g) = e$ for all $g \in G$ this implies that

$$g(g_j H) = g_j H$$

for all $g \in G$. In particular, $gH = H$ for all $g \in G$ which implies $G = H$. This is a contradiction. QED.

EXERCISE 10. The number of conjugacy classes in a group G is called its **class number**. Prove that if G is a finite group with an even class number then $\#G$ is even.

Solution. We shall prove the contrapositive, i.e. we assume that $\#G$ is odd and show that there is an odd number of conjugacy classes. We consider the conjugacy action:

$$\star : G \times G \longrightarrow G, \quad (g, x) \mapsto gxg^{-1}.$$

Then we know that $\#\text{Orb}(x) \cdot \#\text{Stab}(x) = \#G$ which implies that both $\#\text{Orb}(x)$ and $\#\text{Stab}(x)$ are odd for each $x \in G$. Now, using that G is the disjoint union of conjugacy classes (i.e. orbits) it follows that $\#G$ is odd. QED.

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⁹Since H is not assumed to be normal in G we cannot lend this a group structure.